

Cybersecurity Project 2 – Basic Cryptography (Hashing)

Objective

To understand and use basic hashing techniques using the Linux terminal, and learn how hashes can be used for file integrity and password storage.

Tools Needed

- ✓ Kali Linux VM (from your home lab setup)
- ✓ Terminal (pre-installed)

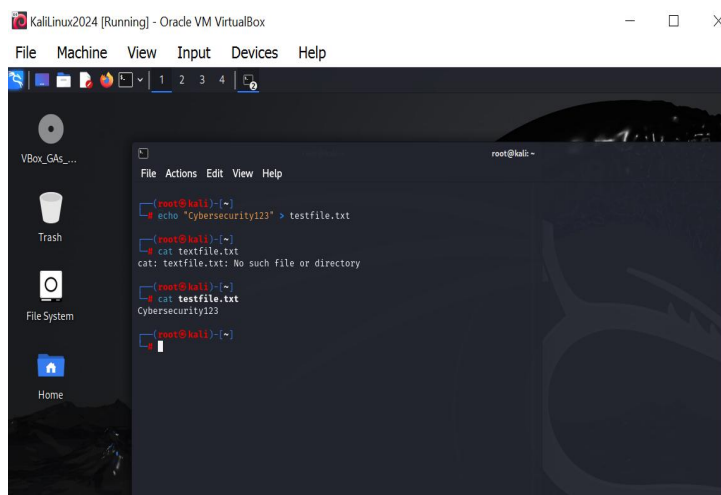
Step 1: Create a Text File

Open the Terminal in Kali.

echo "Cybersecurity123" > testfile.txt

This creates a file called testfile.txt and writes the word Cybersecurity123 into it.

Your terminal after running the command and confirming file creation with: **cat testfile.txt**



Step 2: Generate MD5 Hash

`md5sum testfile.txt`

This gives you an MD5 hash a kind of fingerprint of the file. If the file changes, the hash will change too.

```
(root@kali)~# cat testfile.txt
Cybersecurity123

(root@kali)~# md5sum testfile.txt
ee52f2a4e76732a5e120a6524b7a746e  testfile.txt

(root@kali)~#
```

Step 3: Generate SHA256 Hash

`sha256sum testfile.txt`

This gives you a **stronger** hash (SHA256). Still a fingerprint, but harder to fake or break.

```
(root@kali)~# md5sum testfile.txt
ee52f2a4e76732a5e120a6524b7a746e  testfile.txt

(root@kali)~# sha256sum testfile.txt
74e4f8da17bd81e753ec17c0df8aa6e90a4ceecf85eff6f0984542dd1c7d3794  testfile.txt

(root@kali)~#
```

Step 4: Change the File and Check Again

```
echo "I changed it!" >> testfile.txt
```

Now recheck the hashes:

```
md5sum testfile.txt
```

```
sha256sum testfile.txt
```

You added new text to the file. So the content changed → the hash will also change.



```
(root@kali)~# echo "I changed it" > testfile.txt
(root@kali)~# cat testfile.txt
I changed it
(root@kali)~# md5sum testfile.txt
c9be2472d5e96a82a35179f19bc46841  testfile.txt
(root@kali)~# sha256sum testfile.txt
64aa992e034727db2de33d3a77ccd1943b731f4ae2250d4b786dac3cc8055cbb  testfile.txt
(root@kali)~#
```

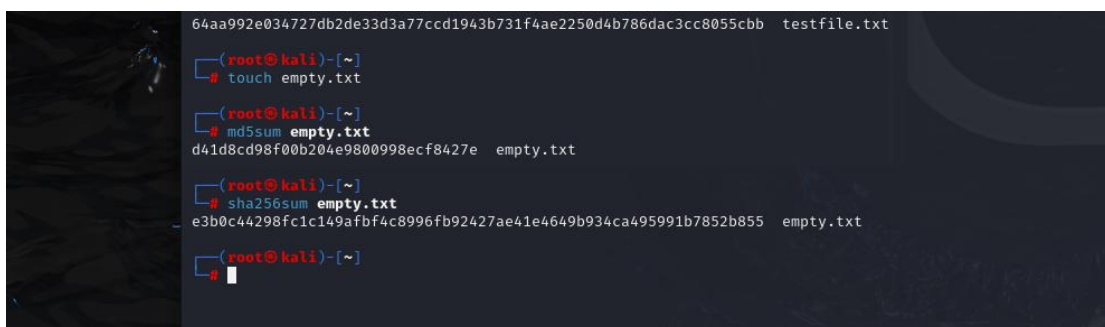
Step 5: (Optional) Try with an Empty File

```
touch empty.txt
```

```
md5sum empty.txt
```

```
sha256sum empty.txt
```

Even an empty file has a unique hash! This shows how everything has a hash.



```
64aa992e034727db2de33d3a77ccd1943b731f4ae2250d4b786dac3cc8055cbb  testfile.txt
(root@kali)~# touch empty.txt
(root@kali)~# md5sum empty.txt
d41d8cd98f00b204e9800998ecf8427e  empty.txt
(root@kali)~# sha256sum empty.txt
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855  empty.txt
(root@kali)~#
```